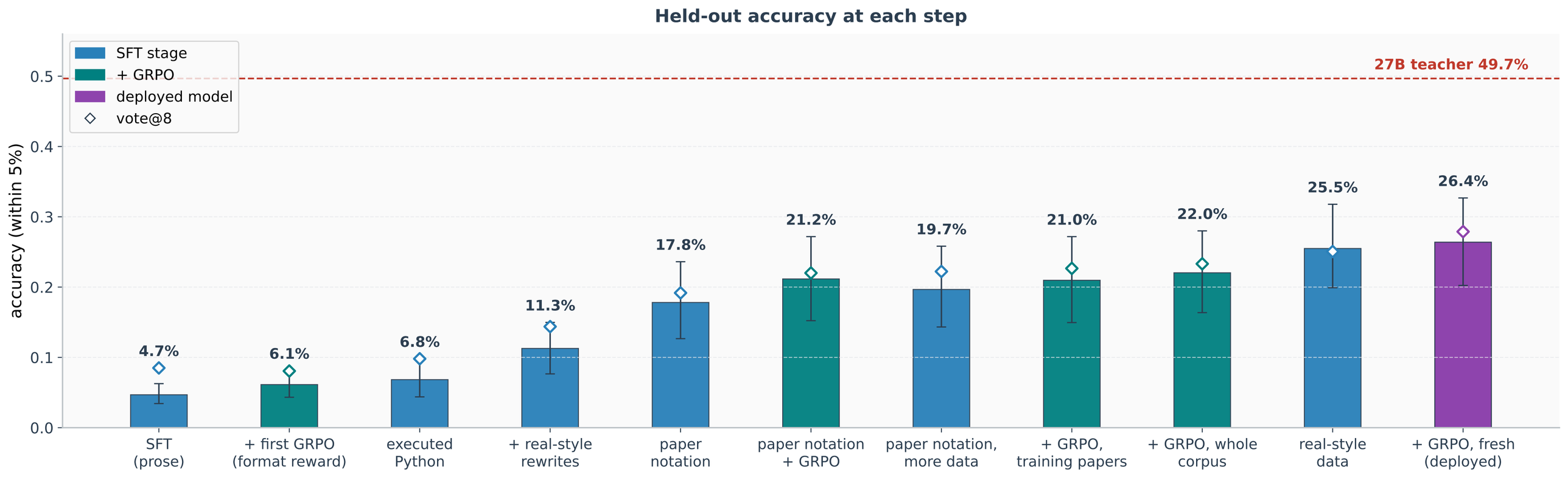
Held-out accuracy at each step
SFT stage
+ GRPO
deployed model
vote@8
27B teacher 49.7%
accuracy (within 5%)
0.0
0.1
0.2
0.3
0.4
0.5
4.7%
6.1%
6.8%
11.3%
17.8%
21.2%
19.7%
21.0%
22.0%
25.5%
26.4%
SFT (prose)
+ first GRPO (format reward)
executed Python
+ real-style rewrites
paper notation
paper notation + GRPO
paper notation, more data
+ GRPO, training papers
+ GRPO, whole corpus
real-style data
+ GRPO, fresh (deployed)